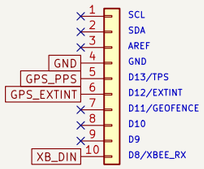
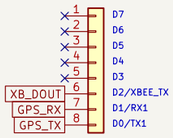
J1



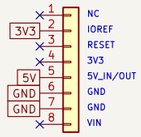
ArduSimple SCL..D8

J2



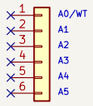
ArduSimple D7..D0

J3



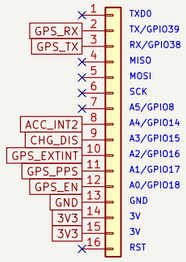
ArduSimple power

J4



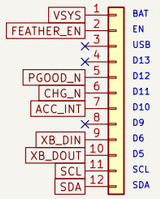
ArduSimple A0..A5

J5



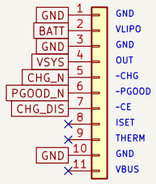
Feather left

J6



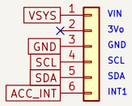
Feather right

J7



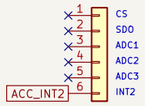
bq24074

J8



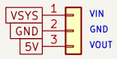
LIS3DH

J9



LIS3DH (support)

J10



S9V11F5

J11

Battery JST-PH

JP1

Charger bypass: shunt only without bq24074

TP1



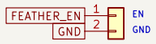
GND

TP2



GPS_EN wire to S9V11F5 EN

J12



Feather off: short EN to GND

Plug-in carrier: every module sits on sockets. Nets join by label name.

ArduSimple: male headers soldered underneath; power switch DOWN (5V IN, IOREF N/C); --role rover (UART2 off).

Feather on 8.5 mm stacking headers (housings on the display side), tails into J5/J6; display up, USB-C at the front. LIS3DH runs from VSYS (its own 3.3 V regulator).

bq24074 optional: fitted -> JP1 open, battery feeds the charger, charger OUT feeds VSYS.

No bq24074 -> fit the JP1 shunt (BATT to VSYS). Never both: the shunt would bypass the charger.

~CHG and ~PGOOD are open drain to Feather D11 / D12 (read with INPUT_PULLUP); the firmware does not use them yet.

Optional, unused by the firmware so far: TIMEPULSE -> A1, EXTINT <- A2, bq ~CE <- A3 (high stops charging; 10k pull-down on the bq), LIS3DH INT2 -> A4.

GPS_EN: Feather A0 to pad TP2; wire TP2 to the S9V11F5 EN hole. Build with -DGPS_POWER_PIN=18 to cut the GPS and XBee in deep sleep.

J12: short EN to GND to switch the Feather off (the regulator and ArduSimple stay on battery power).

Feather S3 Reverse TFT, ArduSimple Budget, LIS3DH, S9V11F5, optional bq24074

race-tracker

Sheet: /

File: esp32-rover-carrier.kicad_sch

Title: ESP32 rover carrier

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 1/1